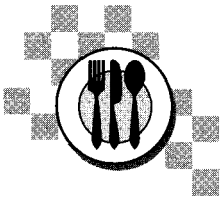
CHAPTER 16



Low-Moisture-Food Sanitation

About This Chapter

In this chapter you will learn:

1. Why good sanitation is important in low-moisture food-processing facilities, even though microbes do not usually grow in these foods
2. About the pests, insects, and other contaminants that affect low-moisture foods
3. How the design of a low-moisture-food plant and the type and layout of equipment affect sanitation
4. Why and how employees should inspect low-moisture foods when they arrive at the plant and while they are in storage
5. How to store low-moisture foods
6. How to clean low-moisture food plants

Introduction

Low-moisture foods include:

- Bakery goods—bread, rolls, muffins, cookies, cakes, pies, pastries, doughnuts, etc.
- Nuts and seeds
- Candy
- Cereal—ready-to-eat cold cereal and cereal to be eaten hot (e.g., oatmeal)
- Grains—flour, cornmeal, rice, dry pasta, baking mixes, etc.
- Snack foods—chips, crackers, pretzels, popcorn, etc.

These foods are called low-moisture foods because they contain very little water, usually less than 25%. This means that most bacteria do not grow in these foods. Even though these foods are less likely to spoil than dairy products, meat, poultry,

fish, fresh fruits and vegetables, and beverages, they can still be contaminated and need hygienic processing and storage.

Just like other food plants, low-moisture-food manufacturing plants need a good sanitation program. The plant must meet the requirements of the U.S. Food and Drug Administration (FDA), the state, and local authorities. Rigid sanitation is needed to make sure that consumers buy safe and wholesome foods.

Bakeries, bottlers, and food warehouses are usually thought of as low risk for food contamination. But the Office of the Inspector General of the Department of Health and Human Services has said that these facilities are becoming more risky because of poor inspection. FDA regulates firms that ship products between states. State and local authorities inspect their facilities, but inspections are rare and look mainly for problems with birds, rodents, and insects. Firms with unsanitary conditions are a public-health risk. Low-moisture-food operations need a good sanitation program to make sure that customers get wholesome foods.

Tidiness and cleanliness help food operations to be efficient, help promote a good image for the company, and affect whether an operation makes a profit or even stays in business. Poor sanitation leads to dissatisfied customers, decreased sales, and a bad reputation for the firm.

FACILITY DESIGN AND CONSTRUCTION

Choosing a Site

A good location for a low-moisture food-processing plant has the following features:

- Level ground or only a slight slope with good drainage
- No springs or areas where water collects
- Available municipal services (sewage, police, and fire)
- Access to major railways and highways in all weather and seasons
- Large enough to double the size of the production facilities and truck parking area
- Not close to areas scheduled for redevelopment or highway construction
- Good supplies of inexpensive and good-quality water, natural gas, electricity, fuel, and other utilities
- Not close to incinerators, sewage treatment plants, and other sources of bad smells, pests, or bacteria
- In an area where gases produced during heating and baking are allowed to be released into the air
- Away from areas that flood, have earthquakes, or have other natural disasters
- Near a good labor market for current and future staff needs

Design of the Outside

The outside of the building should look clean and neat and give a good impression to employees and visitors. The walls should be smooth and waterproof and should

not have ledges or overhangs where birds can live. The builder should seal the walls against rodents and insects. Driveways should be paved and have no weeds growing on them. Trash and water should not collect on driveways. Staff should sweep the outside area regularly so that dust does not blow into storage areas.

Design of the Inside

Walls and framing. Exposed pipes and structures may be fine in areas where food is not processed or stored, as long as they are kept clean and dust-free. Reinforced concrete construction is best for areas where food is processed or stored. The area should have as few inside columns as possible. Staff doors should have hydraulic or spring hinges so they close by themselves, and the doors should have screens. Gaps at the bottom of doors should be less than 0.6 cm (0.25 in), and screens should be at least 20 mesh.

Walls must not have cracks and crevices and must not soak up water or other liquids, so that they can be cleaned easily and thoroughly. Wall finishes should be appropriate for each area. Glazed tile is a good surface for walls in the processing area. Another alternative is fiberglass-reinforced plastic panels, painted with epoxy or coated with other materials that meet company and regulatory standards. Paint is cheap but is not good for food areas, because it cracks, flakes, and chips after a while and needs to be redone often.

In bakery facilities, insulation can be a home for insects. The builder should place insulation on the outside of the building so that pests cannot live in it inside the building.

Ceilings. Suspended ceilings are fine in areas where food is not processed or stored, so long as it is possible to inspect the space above the ceiling and keep pests, dust, and other debris out. The builder must seal the ceiling panels into the grid, but workers must be able to remove them easily for cleaning.

Suspended ceilings are not suitable for food production or handling areas. Suspended ceilings can mold if they get wet and provide a place for pests to live. In areas of bakeries that use flour, dust collects above the ceiling very quickly and can cause fires or explosions, as well as provide food for pests and microbes.

Whenever possible, building designs should not use overhead structures such as joists and beams. Precast-concrete roof panels form a clean, clear ceiling. Precast panels can be made with a smooth inside surface, coated to resist dust, and easily cleaned. Supports for overhead equipment; gas pipes; water, steam, and air lines; and electrical wires should not pass over areas where food is open, clutter the ceiling, and drip dust or water onto people, equipment, or food. A mechanical mezzanine floor can be used for overhead utility equipment. It leaves the ceiling easy to clean and free from horizontal pipes and ducts (see Fig. 16.1).

Floors. Floors in areas that are washed with water must not soak up water, must not have cracks and crevices, or react with chemicals and acids. Yeast foods (e.g., yeast mixtures used to make bread) are very corrosive. Builders must seal floor joints and

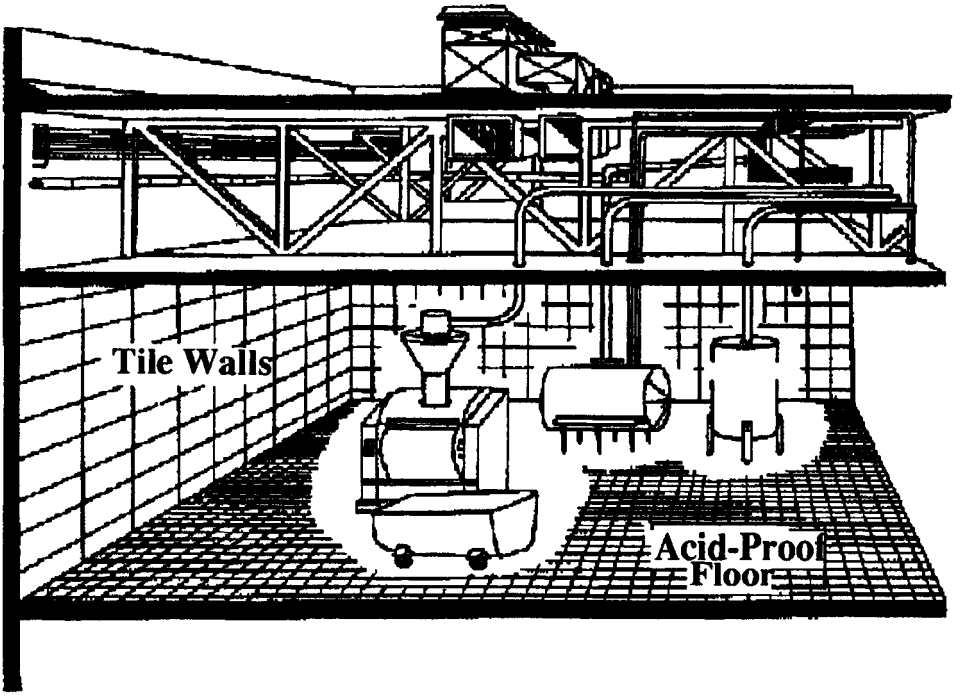


FIGURE 16.1. Specially designed mechanical mezzanine floor separates ducts and support structures from the bakery mixing room. This cuts down on overhead cleaning, makes it easier to get to equipment for maintenance, and protects food products.

cover and seal wall junctions. Whenever possible, builders should use expansive concrete to cut down on the number of joints. In 1986, the U.S. Department of Agriculture recommended that floors should have a slope towards drains with about a 2% grade. This allows proper drainage after wet cleaning. Processing equipment should connect to drain lines, and equipment should have drip pans to cut down on floor spills.

Storage areas should have a white stripe 0.5 meters (18 in) wide painted around the edge. Nothing should be stored in this area. Food must be stored away from non-food items. Some products must be kept separate so that they do not contaminate each other, for example, foods and chemicals.

Different types of operation need different floors. Reinforced concrete, coated or hardened to keep away dust, may be fine for packaging and oven areas of bakeries. But areas used for liquid fermentation and handling dough should have a surface that will not be damaged when it comes in contact with hot water, steam, acids, sugar, and other ingredients or sanitizing chemicals during food processing and cleaning.

Chemical-resistant floors are best for areas that are cleaned using water. Epoxies, polyester, tile, or brick are all good. The builder should bond toppings right onto the base, such as concrete. Toppings give a watertight barrier that protects the concrete.

"Dairy" tile or pavers are good for areas with heavy traffic or for surfaces that touch food products or cleaning solutions. Builders should install dairy tile with acid-resistant bonds. It is very durable and sanitary. It is easy to clean and can be manufactured with a nonslip finish. It is expensive, but it can save money in the long run.

Floors in special areas, such as coolers and freezers, need to be made of the best materials for the area and need good insulation and ventilation. If the freezer floor does not have insulation, it can freeze the ground under it, which can crack or buckle the freezer floor and jam the doors.

Ventilation and Dust Control

It is very important to control dust. The manufacturing process heats the food above pasteurization temperatures and usually kills live microbes. But spores can survive inside baked goods, especially those that are soft and moist. Also, dust from raw materials can contaminate finished food products.

Processors must design facilities so that finished foods are not contaminated. Processors must consider sanitation procedures, the arrangement of the equipment, ventilation, and dust control. They must also control the temperature and humidity so that bacteria cannot grow quickly.

Planning Equipment

The type of equipment and how it is arranged affect productivity. A mechanical mezzanine floor should separate heating and cooling equipment from processing areas. The plant should have high-efficiency motors and electrical equipment (Fig. 16.1), the best automatic technology and controls that the processor can afford, and a flexible design to respond to changes in processing and consumer demands. All equipment should meet current regulations.

Sanitation

Like other food-processing facilities, the building is there to protect what is inside. The building must protect the process, equipment, and food products. Sanitation, productivity, product flow, and choice of equipment and site are all very important.

An operation such as breadmaking involves fermentation. Fermentation uses yeast, and the bakery must be kept sanitary for the right organisms to grow. Poor sanitation allows natural microorganisms to ferment the dough instead of the proper yeast. Once they are growing in the facility, natural microorganisms are very difficult to remove and difficult to control.

Current Good Manufacturing Practices (CGMPs) related to design and construction should:

1. Allow enough space for equipment and storage
2. Separate operations that might contaminate food

3. Provide enough light
4. Provide enough ventilation
5. Protect against pests

It is best to keep the inside of the plant simple and uncluttered. Designers should design the plant layout with sanitation in mind. The design team should include technical staff, engineers, and production staff from the plant, as well as a contracted design and engineering firm.

Productivity

The well-designed, efficient plant makes a productive operation. A good design should have:

- Flexibility to allow for future changes
- Enough space to process a larger volume to allow for growth
- Dry and refrigerated storage areas in convenient places
- A central plant management office and laboratory so that managers can supervise all processes closely
- Ingredient storage areas near where they are mixed and used
- Secondary equipment, such as boilers and refrigeration machines, in places that minimize pipes and utilities
- Open space around and under equipment for cleaning and maintenance
- Manufacturing equipment arranged for easy cleaning in place (CIP)
- Proven or tested equipment and processes
- State-of-the-art controls and automation; facilities should update systems as technology advances and becomes less expensive
- Conveniently located restrooms, lockers, and break areas away from work areas
- Plenty of handwashing stations in work areas
- A computer simulation of the proposed layout, if possible
- Extensive design input from the plant's engineering and operation staff to make sure it complies with the company's standards and will not cause problems in day-to-day operations
- Compliance with federal, state, and local regulations

RECEIVING AND STORING RAW MATERIALS

Checking Raw Materials

Staff cannot check all of the raw materials that come into the plant. They need a system for collecting samples to decide if they should accept or reject deliveries. They need to collect enough samples (a statistically valid size and number) to be confident that the raw materials are acceptable.

Inspecting Trucks

Trucks used to transport low-moisture raw materials before, during, and after unloading should be inspected. Inspectors should look at the overall condition of the vehicle, areas where food and dust can collect, and potential homes for insects. It is important to check for insects in the area around doors or hatches. Inspectors should look for crawling or flying insects, animal tracks, nesting materials, smells, and feces. Pellets and smells are signs of rodents; feathers or droppings are signs of birds.

Evaluating Raw Materials

Inspectors should check the moisture content of raw materials. They can analyze for percentage moisture, but can also check for signs of high moisture content. A sour or musty smell can be caused by mold, which can only grow on products such as cereal grains when they are moist. If inspectors think the material may be moldy, they should do more tests. Cereal grains with more than 15% moisture should not be stored for long because they are likely to mold and attract insects.

Inspectors should also check that raw materials do not smell of pesticides that may have been used to get rid of insects. The inspector may need to test the food to find out if the pesticide levels are too high.

Inspectors should check samples of raw materials to see what percentage, if any, has been destroyed by insects or rodents. They should also check the amount of dust, other contaminants, insect webs, molds, smells, live and dead insects, and rodent droppings or hair. Immature insects may be inside grains; X-ray machines or cracking-flotation methods can test for these.

Inspecting goods when they arrive reduces pest damage and helps prevent raw materials from contaminating the end product. Pests can "hitchhike" into buildings and settle on ingredients, packages, pallets, and machinery, so inspectors need to check all of these. A food processor has the right to reject any materials that come to the plant or hold them for more tests. Trained and experienced personnel should decide to accept or reject shipments.

Storing Food

Many low-moisture-food-processing plants store material such as grain before processing it. The United States stores more grain than any other commodity. But grain contains mold spores and insect eggs that grow and damage the grain under some conditions. If grain kernels are physically damaged, insects and molds can get into them and spoil them more easily. Insects can also cause biological damage to kernels, which allows molds to get inside.

If a processor is going to store grain for more than 1 month, the grain needs special treatment:

- Inspectors should make sure that it is not already infested or infected.
- Moisture content must be less than 13.5%.

- Processors may clean grain before storage, using aspiration or other methods to remove insects, weed seeds, and other foreign materials.
- Processors may use chemicals to protect the grain from insects.
- Processors may fumigate grain using a modified atmosphere such as carbon dioxide and nitrogen. Fumigation using these inert (nonreactive) gases is popular because chemical use is very restricted. Inert gases do not leave residues, but a storage bin that contains an inert gas can be as deadly to human beings as a lethal chemical, because inert gas does not contain any oxygen.
- Aeration systems can reduce insect feeding and growth in storage bins.

Processors should control dust in handling and storage areas. Keeping dust away saves time spent cleaning floors, walls, ledges, and equipment. Suction (reduced pressure) reduces dust on grain conveyors, receiving hoppers, bucket elevators, bins, and places where grain is transferred from one piece of equipment to another.

Adding very pure oils to grain as it goes into storage can reduce dust. Processors should add the oil (up to 200 ppm) as soon as possible after the grain comes off the truck to reduce dust throughout the plant.

Processors still need to control storage conditions of root crops, such as potatoes, to prevent *Fusarium* tuber rot and bacterial soft rots. The potato industry uses well-ventilated storage rooms with concrete floors to control these problems. Of the potatoes harvested each year, 80 to 90% are stored in good facilities.

Processors usually store bulk oils and shortenings in large carbon or stainless-steel tanks. Staff need to wash these containers with a strong alkaline solution or alkali plus detergents to keep them clean. Nitrogen "blankets" over oils help protect them. Processors need to make sure that oils are not splashed and shaken too much during bottling, because this can cause oxidation, which makes the oil rancid. Processors should coat clean bulk tanks (especially carbon steel tanks) with oil to seal them and prevent rust.

Housekeeping in Raw-Food Storage Areas

Low-moisture food storage areas need good maintenance and housekeeping to keep them sanitary. Processors should maintain bulk storage areas so that they do not have cracks or ledges that can collect dust, dirt, debris, microbes, and insects. Staff should inspect empty bins or other containers to make sure they are in good repair and do not still contain bits of food. Even small amounts of food can allow insects to grow.

Inspectors should look for insect tracks in the dust on floors and walls, and check for moths. They should check for dust in damp areas, where molds, mites, and fungus-feeding insects can grow. Inspectors should also check for unusual smells that could come from mold, insects, or chemicals. It is important to inspect elevators and conveyors that may hold food residues. Equipment that is not being used may hold residues that insects can grow on. Once insects are established in the building, it is easy for them to move in food storage areas.

Staff need to check regularly that storage areas do not have indications of live insects on surfaces, floors, and walls. When processors are going to store grain for a

long time, they may use thermocouple cables to monitor the temperature. If the temperature goes up, it may be because insects or molds are growing in the grain. Mold will not grow if the grain is kept dry. If inspectors find insects in the grain, they should treat it or fumigate it. Heat from insects can make moisture spread, which allows mold to grow. Inspectors should keep thorough records of all checks, tests, cleaning, fumigation, or other steps.

Processors should keep storage areas organized so that they can inspect and clean them easily. Well-organized storage areas are less likely to have sanitation problems in the first place. Inspectors and other employees must know how to look for pests and how to get rid of them.

Recommendations for storage include:

- Stacking bags and cartons on pallets
- Spacing bags and cartons away from the walls and from each other for easy inspection and cleaning
- Rotating stock to give insects and rodents less time to get into foods
- Inspecting the area using a flashlight to see into dark corners, under pallets, and between stacks
- Looking for insects that are flying; crawling on walls, ceilings, and floors; and hovering over bags and cartons
- Sifting food to check for insects; (Chap. 9 gives information on controlling insects and rodents.)

Processors should decide how often to clean and inspect storage areas based on the temperature and humidity. At room temperature (25–30°C, 77–86°F), the life cycle of many insects that live in low-moisture grains and foods is about 30 to 35 days. Insects usually stop reproducing below 10°C (50°F). Thus, storage areas should be cleaned and inspected more often when they are warm than when they are cool. The temperature of the raw food affects insects more than the air temperature in the area. Humid areas should be cleaned and inspected more often than dry areas, because moisture allows molds, yeast, and bacteria to grow. Good ventilation and suction can help reduce humidity.

Superheating can kill pests in dry storage areas. But it takes a lot of heat to kill insects, so this process uses a lot of energy, especially during cold weather. Portable heating units can superheat an individual piece of infested equipment. Designs for new or renovated facilities should include equipment for superheating. Keeping storage areas cold enough to stop insects from breeding is not practical, because refrigeration is expensive, and freezing can damage equipment or the facility.

Inspecting Storage Areas

Inspections and written reports of results should be done. The plant should have an inspection report form with a scoring system. The form should describe what each score means. Plants and warehouses may use three rating levels:

- *Acceptable,* if it meets most of the requirements

- *Provisionally acceptable,* if it needs corrective steps to meet the standards
- *Unacceptable,* if it does not meet the standards and the operation is unsanitary

Inspection in processing and storage areas should check for conditions that could contaminate food and correct them before they become a problem. Low-moisture foods have a lower water activity (A_w), so microbes are less of a problem. Other contaminants need more attention in low-moisture-food processing.

Inspectors should check overhead areas for flaking paint, conditions that block cleaning, dust, and condensed water. Inspectors should look for broken window panes and damaged or missing screens at ground level, in the basement, and above ground level. Inspectors should report open windows or other places where pests could get in. They should check for signs of pests, such as insect trails in dust, rodent droppings, and bird droppings or feathers. Employees should report signs of pests so that management can find the source of the problem and correct it. All employees should know how to look for signs of pests.

Operations staff should check the outside of equipment and overhead equipment. Maintenance and sanitation staff should check the inside of equipment to make sure it is sanitary. Some equipment has dead spots where food can collect. Staff need to check these parts regularly when the equipment is not in use. Processors should use equipment with easy access to the inside through cleanout openings or equipment that is easy to take apart. Staff should remove equipment from the area when they are not using it, if possible, and should leave it open so that food particles go though it and can be seen easily.

CLEANING PROCESSING PLANTS

The manufacturing area of low-moisture-food plants should be cleaned every day. Some cleaning should be done while the plant is operating to make sure that it stays tidy. But some cleaning steps and most equipment cleaning (especially the insides) have to wait until production stops. Employees can combine some of the cleaning with routine maintenance. If equipment is easy to reach, employees are more likely to clean it properly and prevent infestation.

Most cleaning equipment is easy to use. Hand brooms, push brooms, and dust and wet mops are the basic cleaning tools. Brushes, brooms, and dustpans remove heavy debris and work well on semismooth floors. Dust mops are faster for cleaning smooth floors with small amounts of dust. Vacuum cleaning is the best way to clean equipment in many production areas. It is thorough and removes light and moderate debris from smooth and rough surfaces. Vacuum cleaners hold dust and do not need a secondary way to collect it (in the way that a broom needs a dustpan). Smaller operations can use portable vacuum cleaners. Larger facilities may need a central (installed) vacuum system. Installed equipment is more expensive but may be more convenient. In large storage areas with nonporous floors, staff may use a mechanical scrubber or sweeper to keep the floor sanitary.

Facilities may have a compressed air line to remove debris from equipment and other difficult-to-reach areas. This is an easy way to clean hard-to-reach areas and is

safer than having employees climb ladders with brushes. But compressed air spreads dust from one place over the whole area, so it may spread infestation. Compressed air should have low volume and low pressure so that it does not disperse dust too much. Employees who use compressed air should wear safety equipment such as dust respirators and safety goggles.

Some equipment needs special cleaning tools. Cylindrical brushes are good for spouts. Workers can drag brushes through spouts using rope or cords or use motorized brushes.

Equipment must be well-organized and properly installed so that staff can clean the equipment and the area around it easily. The facility should have specific storage areas for ingredients and supplies. Supervisors should make sure trashcans for bags, film, paper, and waste products are in convenient places in manufacturing, packaging, and shipping areas.

SUMMARY

- Strict sanitation is essential in low-moisture-food-processing and storage facilities. Sanitary facilities make safe food products and meet regulations.
- It is important to choose a good site and a good building design for the plant.
- Processors should choose equipment carefully and design the layout so that the equipment is easy to clean and inspect.
- Samples of raw foods should be collected when they are delivered to make sure they are not infested with pests, molds, rodents, or other contaminants.
- Good housekeeping protects raw and processed foods during storage.
- Storage areas should be inspected often to check for microbes and pests.
- Frequency of inspection and cleaning of storage areas depends on the temperature and humidity.
- The manufacturing area should be cleaned every day.
- In low-moisture product areas, basic cleaning tools can be used, such as vacuum cleaners, powered floor sweepers and scrubbers, and compressed air.

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STUDY QUESTIONS

1. What are low-moisture foods?
2. List three things that often spoil low-moisture foods.
3. Why is it important to control dust in low-moisture-food-processing plants?
4. What can processors do to prevent and control infestation of low-moisture foods?
5. What are the advantages and disadvantages of superheating grain?
6. List six types of cleaning equipment used in low-moisture-food-processing plants.

TO FIND OUT MORE ABOUT LOW-MOISTURE-FOOD SANITATION

1. Contact the Pillsbury Company, Pillsbury Center, Minneapolis, MN 55402, and ask for a copy of their leaflet "Preventing or Eliminating Insects in Grain-Based Foods" and other food safety resources.
2. Contact a local bakery or snack food manufacturer, and ask how they keep insects, pests, and mold under control.
3. Put four slices of bread from the same loaf in four plastic bags. Add a tablespoon of water (to add moisture or humidity) to two of the bags. Seal the bags. Place one moist and one dry bag in the refrigerator. Place one moist and one dry bag on the counter. Notice how many days it takes for each slice of bread to show signs of mold. The moist slice of bread at room temperature should grow mold before the others.